15. Rectifiable sets and approximate tangent planes

Rectifiability is one of the most fundamental concepts of geometric measure theory. Familiar examples of rectifiable sets are rectifiable curves and m-dimensional C^1 submanifolds of $\mathbf{R}^n$. But for various reasons one should allow more complicated sets to be called in some sense rectifiable. One such reason is the desire to formulate and prove analogues of classical geometric results in their greatest possible generality. Also solutions to many natural geometric variational problems are often rectifiable but not smooth. A third reason is the powerful compactness theorems for rectifiable surfaces which can only hold if we do not restrict ourselves to a too narrow class. This can be formulated vaguely to mean that anything that can be approximated in a sufficiently strong sense by rectifiable sets should also be called rectifiable. Let us look at this statement in the light of two examples.

Two examples

15.1. Example. Let $q_1, q_2, \dots$ be the points with rational coordinates in the unit disc $B = \{x \in \mathbf{R}^2 : |x| \leq 1\}$. Let

$$E = \bigcup_{i=1}^{\infty} S_i \quad \text{where } S_i = \{x \in \mathbf{R}^2 : |x - q_i| = 2^{-i}\}.$$

Then $\mathcal{H}^1(E) \leq \sum_{i=1}^{\infty} 2\pi \cdot 2^{-i} = 2\pi$ (if fact, $\mathcal{H}^1(E) = 2\pi$, as one easily sees). The finite unions of circles $\bigcup_{i=1}^{k} S_i$ should certainly be considered as rectifiable, if we are to allow something more general than rectifiable curves. Now E can be approximated from inside by these finite unions in measure:

$$\lim_{k\to\infty} \mathcal{H}^1\Big(E \setminus \bigcup_{i=1}^{k} S_i\Big) = 0.$$

Is this approximation sufficiently strong so that E should also be called rectifiable? At first glance it seems not. After all E is dense in B, $\overline{E} \cap B = B$, and so it seems to lose most of the nice properties of rectifiable curves. For example, it does not have a tangent at any point. However, it has turned out that it is exactly this kind of approximation that should preserve the rectifiability. Fortunately it has also turned out that this approximation preserves most of the nice properties of rectifiable curves if we interpret them correctly. Let us look at what this means about the tangents of E.

For E to have a tangent line L in the usual sense at the point a would mean that for any $\alpha > 0$ the two-sided angular sector $S(a, L, \alpha) = \{x : d(x, L) \leq \alpha|x-a|\}$ would contain all points of $E \cap B(a, r)$ for sufficiently small r. We want to study tangents in a measure-theoretic sense. Thus we may ignore sets of measure zero and sets with sufficiently small measure. A convenient way of expressing this for E is that instead of requiring $E \cap B(a, r) \setminus S(a, L, \alpha)$ to be empty, we only require that

$$\lim_{r \downarrow 0} r^{-1} \mathcal{H}^1\big(E \cap B(a, r) \setminus S(a, L, \alpha)\big) = 0.$$

If this holds for all $\alpha > 0$, we call L an approximate tangent for E.

It can now very quickly be seen that E has an approximate tangent at $\mathcal{H}^1$ almost all of its points. It is sufficient to verify this for points on a fixed circle S_i. Because of the upper density theorem 6.2 (2) applied to $A = \bigcup_{j \neq i} S_j \setminus S_i$, we have for $\mathcal{H}^1$ almost all a in $\mathbf{R}^2 \setminus A$, and thus in S_i,

$$\lim_{r \downarrow 0} r^{-1} \mathcal{H}^1\Big(B(a, r) \cap \Big(\bigcup_{j \neq i} S_j \setminus S_i\Big)\Big) = 0.$$

This gives readily that the ordinary tangent of S_i is an approximate tangent of E.

Let us consider another example.

15.2. Example. Consider the self-similar set $F = C(1/4) \times C(1/4)$ where $C(1/4)$ is the Cantor set of 4.10. Then $0 < \mathcal{H}^1(F) < \infty$ and F can also be approximated with nice rectifiable sets. At the k-th stage of the construction, F is contained in 4^k squares of side-length 4^{-k}, thus the union P_k of their boundaries has length 4 and is within distance 4^{-k} from F. Hence F can be approximated in the Hausdorff distance, recall 4.13, by rectifiable sets with uniformly bounded $\mathcal{H}^1$ measure. Also $\mathcal{H}^1 \llcorner P_k \to c\mathcal{H}^1 \llcorner F$ weakly. However this approximation is too weak to preserve any reasonable concept of rectifiability. This is reflected in the tangential properties of F: at none of its points does F possess an approximate tangent. We leave the verification as an exercise.

m-rectifiable sets

Recall that $\Gamma \subset \mathbf{R}^n$ is a rectifiable curve if and only if $\Gamma = f(I)$ for some Lipschitz map f from a bounded interval $I \subset \mathbf{R}$ into $\mathbf{R}^n$. As a generalization we now define rectifiable sets.

Throughout the rest of this chapter m and n will be a positive integers.

15.3. Definition. *A set $E \subset \mathbf{R}^n$ is called m-rectifiable if there exist Lipschitz maps $f_i\colon \mathbf{R}^m \to \mathbf{R}^n$, $i = 1, 2, \dots$, such that*

$$\mathcal{H}^m\Big(E \setminus \bigcup_{i=1}^{\infty} f_i(\mathbf{R}^m)\Big) = 0.$$

A set $F \subset \mathbf{R}^n$ is called purely m-unrectifiable if $\mathcal{H}^m(E \cap F) = 0$ for every m-rectifiable set E.

Note that we are not requiring an m-rectifiable set to be of finite $\mathcal{H}^m$ measure.

It is clear that the set E in Example 15.1 is 1-rectifiable. It is not as immediate, but neither is it very difficult, to prove that the set F in Example 15.2 is purely 1-unrectifiable. Clearly $\mathbf{R}^n$ and all of its subsets are n-rectifiable and also trivially m-rectifiable if $m > n$. The 0-rectifiable sets are exactly the countable sets. Thus we could usually assume $0 < m < n$.

We have adopted the terminology of Federer [3] in an abbreviated form. Federer uses the terms countably $(\mathcal{H}^m, m)$ rectifiable and purely $(\mathcal{H}^m, m)$ unrectifiable. These concepts were first introduced for one-dimensional sets in $\mathbf{R}^2$ by Besicovitch [1]. He took a different starting point and used the terms regular and irregular. Much of Besicovitch's theory is treated in the books of Falconer [4], [16] and the higher dimensional theory extensively in Federer [3] and parts of it in L. Simon [1]. The theory was mainly developed by Besicovitch [1], [4], [5] for $m = 1$, $n = 2$ and by Federer [1] for general m and n.

We shall first discuss some rather immediate consequences of the definitions. The first lemma follows easily from the extension theorem 7.2 of Lipschitz maps.

15.4. Lemma. *A set $E \subset \mathbf{R}^n$ is m-rectifiable if and only if there exist subsets $A_1, A_2, \dots$ of $\mathbf{R}^m$ and Lipschitz maps $f_i\colon A_i \to \mathbf{R}^n$, $i = 1, 2, \dots$, such that $\mathcal{H}^m(E \setminus \bigcup_{i=1}^{\infty} f_i(A_i)) = 0$.*

We leave the simple proof of the following lemma as an exercise.

15.5. Lemma. (1) *Every m-rectifiable set has σ-finite $\mathcal{H}^m$ measure.*

(2) *Any subset of an m-rectifiable set is m-rectifiable.*

(3) *The countable union of m-rectifiable sets is m-rectifiable.*

(4) *If E is m-rectifiable, there is an m-rectifiable Borel set B such that $E \subset B$ and $\mathcal{H}^m(B) = \mathcal{H}^m(E)$.*

Next we shall show that any subset A of $\mathbf{R}^n$ with $\mathcal{H}^m(A) < \infty$ can be decomposed into an m-rectifiable and a purely m-unrectifiable part. Thus the study of the structure of sets with finite $\mathcal{H}^m$ measure is reduced to the study of m-rectifiable and purely m-unrectifiable sets. This will be the main theme in the rest of the book.

15.6. Theorem. *If $A \subset \mathbf{R}^n$ with $\mathcal{H}^m(A) < \infty$ there is an m-rectifiable Borel set B such that $A \setminus B$ is purely m-unrectifiable. Thus A has a decomposition into m-rectifiable and purely m-unrectifiable subsets E and F:*

$$A = E \cup F, \quad E = A \cap B, \quad F = A \setminus B.$$

Clearly the above decomposition is unique up to $\mathcal{H}^m$ null-sets.

Proof. Let M be the supremum of the numbers $\mathcal{H}^m(A \cap B)$ where B ranges over all m-rectifiable Borel subsets of $\mathbf{R}^n$. We can choose for every $j = 1, 2, \ldots$ an m-rectifiable Borel set B_j such that $\mathcal{H}^m(A \cap B_j) > M - 1/j$. Then $B = \bigcup_{j=1}^{\infty} B_j$ is the desired set. □

Linear approximation properties

We shall now study relations between rectifiability and existence of tangent planes. But first we shall formulate closely related approximation properties which will be useful later on. For an affine m-plane W to be a tangent plane (in a measure-theoretic sense to be defined in 15.17) for a set E at a point a means that $W \cap B(a, r)$ should approximate $E \cap B(a, r)$ reasonably well for small r. Conditions (15.8) and (15.9) below give a meaning for this. According to (15.9) most of E lies near W in $B(a, r)$ and (15.8) says that there are no big holes in E near $W \cap B(a, r)$. Definitions 15.7 and 15.10 are introduced mainly for technical reasons. In particular, we shall prove in the next chapter that the weak approximability of 15.10 implies rectifiability and this will provide a substantial part for the proof of the main theorems of Chapter 17. In terms of the tangent measures of $\mathcal{H}^m \llcorner E$ the m-linear approximability corresponds essentially to the condition 16.5 (2) and the weak m-linear approximability to 16.5 (3).

15.7. Definition. *We say that a subset E of $\mathbf{R}^n$ is m-linearly approximable if for $\mathcal{H}^m$ almost all $a \in E$ the following holds: if η is a positive number, there are positive numbers r_0 and λ and an affine m-plane $W \in A(n,m)$ such that $a \in W$ and for any $0 < r < r_0$,*

$$\mathcal{H}^m\big(E \cap B(x, \eta r)\big) \geq \lambda r^m \quad \text{for } x \in W \cap B(a,r), \tag{15.8}$$

and

$$\mathcal{H}^m\big(E \cap B(a,r) \setminus W(\eta r)\big) < \eta r^m. \tag{15.9}$$

Recall that $W(\delta) = \{x : d(x,W) \leq \delta\}$. In the next chapter we shall use the weaker form of this definition where W is allowed to depend on r.

15.10. Definition. *A subset E of $\mathbf{R}^n$ is weakly m-linearly approximable if for $\mathcal{H}^m$ almost all $a \in E$ the following holds: if η is a positive number, there are positive numbers r_0 and λ such that for any $0 < r < r_0$ there is $W \in A(n,m)$ such that $a \in W$ and (15.8) and (15.9) hold.*

Clearly these conditions imply $\Theta^m_*(E,a) > 0$ for $\mathcal{H}^m$ almost all $a \in E$. Observe also that, if $\mathcal{H}^m(E) < \infty$, both types of approximation properties are preserved for $\mathcal{H}^m$ measurable subsets because of Theorem 6.2 (2).

15.11. Theorem. *If E is an $\mathcal{H}^m$ measurable m-rectifiable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$, then E is m-linearly approximable.*

Proof. Let $0 < \eta < 1$, let $f\colon \mathbf{R}^m \to \mathbf{R}^n$ be a Lipschitz map with $L = \mathrm{Lip}(f)$, and let B be an $\mathcal{L}^m$ measurable subset of $\mathbf{R}^m$ with $fB \subset E$. We need to verify the properties (15.8) and (15.9) at $\mathcal{H}^m$ almost all points of fB.

By Theorem 7.9, $\Theta^m_*(fB,a) > 0$ for $\mathcal{H}^m$ almost all $a \in fB$. Hence we may assume that there are $r_0 > 0$ and $\lambda > 0$ such that

$$\mathcal{H}^m\big(E \cap B(a,r)\big) \geq \lambda r^m \quad \text{for } a \in fB,\ 0 < r < r_0; \tag{1}$$

the original B is up to a set of $\mathcal{L}^m$ measure zero a countable union of such subsets, and we may consider each of them separately.

If f is differentiable at a point x, write $L_x = f'(x) - f'(x)x + f(x)$ and $W_x = L_x\mathbf{R}^m$. Further, if $\dim W_x = m$, let $\ell(x)$ be the smallest number

δ such that $|L_x y - L_x x| \geq \delta |y-x|$ for $y \in \mathbf{R}^m$. By Theorems 7.3, 7.5 and 7.6 we may assume $\ell(x) > 0$ to exist on B.

Let $\varepsilon > 0$. Using what was said above and Lebesgue's density theorem 2.14 (1), we can find a compact subset C of B and positive numbers r_0 and $\delta < \min\{\eta/4, 1/L\}$ such that $\mathcal{L}^m(B \setminus C) < \varepsilon$ and that for $x \in C$,

$$|f(y) - L_x y| < \delta^2 |x-y| \quad \text{for } y \in B(x, r_0), \tag{2}$$

$$\ell(x) \geq 2\delta, \tag{3}$$

$$d(y, B) < \delta^2 r \quad \text{for } y \in B(x, r/\delta),\ 0 < r < r_0. \tag{4}$$

We partition C into finitely many Borel subsets C_i such that $d(C_i) < r_0$.

Fix i and let $x \in C_i$, $a = f(x)$, be such that $\Theta^m(E \setminus fC_i, a) = 0$. By Theorems 7.5 and 6.2 (2) it suffices to verify (15.8) and (15.9) at such a point a. Let $0 < r < \delta r_0/2$ and $b \in W_x \cap B(a,r)$, $b = L_x y$. Then by (3), $y \in B(x, r/\delta)$. By (4) there is $z \in B$ with $|y - z| < \delta^2 r$, whence $|x - z| < 2r/\delta$. Then by (2) and the fact that $\|f'(x)\| \leq L < 1/\delta$,

$$\begin{aligned} |f(z) - b| &\leq |f(z) - L_x z| + |L_x z - L_x y| \\ &< \delta^2 |x - z| + L|z - y| < 3\delta r. \end{aligned}$$

Thus, as $4\delta < \eta$, we have by (1)

$$\mathcal{H}^m\big(E \cap B(b, \eta r)\big) \geq \mathcal{H}^m\big(E \cap B(f(z), \delta r)\big) \geq \lambda \delta^m r^m,$$

and we have verified (15.8).

We obtain from (2) that

$$f\big(C_i \cap B(x, r/\delta)\big) \subset W_x(\delta r) \subset W_x(\eta r),$$

and from (3) and (2) using the fact $d(C_i) < r_0$ that

$$f\big(C_i \setminus B(x, r/\delta)\big) \subset \mathbf{R}^n \setminus B(a, r),$$

because for $y \in C_i \setminus B(x, r/\delta)$,

$$\begin{aligned} |a - f(y)| &\geq |L_x x - L_x y| - |L_x y - f(y)| \\ &\geq 2\delta|x - y| - \delta^2|x - y| \geq \delta|x - y| > r. \end{aligned}$$

Thus

$$B(a, r) \cap fC_i \subset W_x(\eta r).$$

Since $\Theta^m(E \setminus fC_i, a) = 0$, (15.9) follows. $\square$

Rectifiability and measures in cones

It would now follow very quickly that an $\mathcal{H}^m$ measurable m-rectifiable set with $\mathcal{H}^m(E) < \infty$ has an approximate tangent plane at $\mathcal{H}^m$ almost all of its points. But before stating the theorem or defining the approximate tangent planes we start to prove that the converse also holds. For this we recall some notation from Chapters 3 and 11.

15.12. Notation. *Let $V \in G(n, n-m)$ be an $(n-m)$-plane through the origin. Recall that $P_V : \mathbf{R}^n \to V$ is orthogonal projection. Let*

$$Q_V = P_{V^\perp} : \mathbf{R}^n \to V^\perp$$

be orthogonal projection onto the orthogonal complement of V (that is, $P_V + Q_V =$ identity map). If $a \in \mathbf{R}^n$, $0 < s < 1$ and $0 < r < \infty$ we set

$$\begin{aligned} X(a, V, s) &= \{x \in \mathbf{R}^n : d(x-a, V) < s|x-a|\} \\ &= \{x \in \mathbf{R}^n : |Q_V(x-a)| < s|x-a|\} \end{aligned}$$

and

$$X(a, r, V, s) = X(a, V, s) \cap B(a, r).$$

The following lemma is simple but very important. In fact, in all the proofs below where Lipschitz maps are needed to show rectifiability, the Lipschitz maps can be produced with the aid of this lemma.

15.13. Lemma. *Suppose $E \subset \mathbf{R}^n$, $V \in G(n, n-m)$, $0 < s < 1$, and $0 < r < \infty$. If*

$$E \cap X(a, r, V, s) = \emptyset \quad \textit{for all } a \in E,$$

then E is m-rectifiable.

Proof. Since E is a countable union of sets whose diameters are less than r, we may assume $d(E) < r$. Let $a \in E$. If $|Q_V a - Q_V b| < s|a-b|$ and $|a-b| \le r$, then $b \in X(a, r, V, s)$, and so $b \notin E$ by the hypothesis. This means that $|Q_V a - Q_V b| \ge s|a-b|$ for a, $b \in E$. Hence $Q_V|E$ is one-to-one with Lipschitz inverse $f = (Q_V|E)^{-1}$, $\mathrm{Lip}(f) \le 1/s$. Since $Q_V E$ lies on an m-plane and $E = f(Q_V E)$, E is m-rectifiable. □

We stated this lemma in its simplest form which will be sufficient for us. However, expressing E suitably as a countable union, one can easily show that V, s and r can be allowed to depend on a. Next we take a more essential step, further replacing the emptiness of the intersection by a bound on its measure.

https://doi.org/10.1017/CBO9780511623813.017 Published online by Cambridge University Press

15.14. Lemma. *Let $V \in G(n, n-m)$, $0 < s < 1$, $0 < \delta < \infty$, and $0 < \lambda < \infty$. If $A \subset \mathbf{R}^n$ is purely m-unrectifiable and*

$$\mathcal{H}^m\big(A \cap X(x,r,V,s)\big) \le \lambda r^m s^m \quad \text{for } x \in A,\ 0 < r < \delta,$$

then

$$\mathcal{H}^m\big(A \cap B(a,\delta/6)\big) \le 2 \cdot 20^m \lambda \delta^m \quad \text{for all } a \in \mathbf{R}^n.$$

Proof. We may assume $A \subset B(a, \delta/6)$ and

$$A \cap X(x, V, s/4) \neq \emptyset \quad \text{for } x \in A,$$

because the subset of A where this fails has zero $\mathcal{H}^m$ measure by Lemma 15.13. Let

$$h(x) = \sup\big\{|y - x| : y \in A \cap X(x, V, s/4)\big\}, \quad x \in A.$$

Then $0 < h(x) \le \delta/3$. Choose

$$x^* \in A \cap X(x, V, s/4) \quad \text{with } |x - x^*| \ge 3h(x)/4.$$

Letting

$$C_x = Q_V^{-1}\big[Q_V B(x, sh(x)/4)\big],$$

we have

(1) $\quad A \cap C_x \subset X(x, 2h(x), V, s) \cup X(x^*, 2h(x), V, s) \quad \text{for } x \in A.$

It is not hard to convince oneself about this geometrically (see Figure 15.1) but let us prove it rigorously.

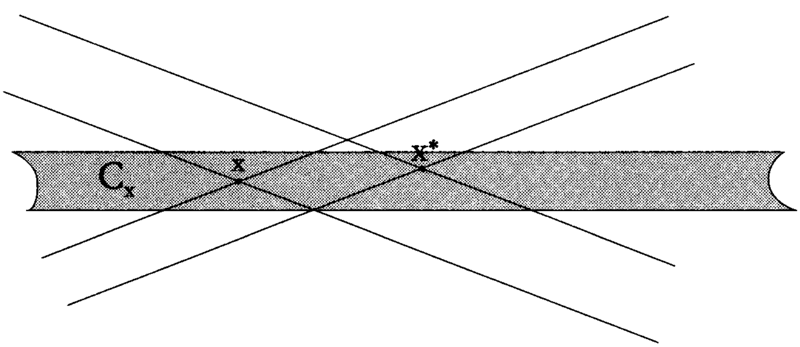

Figure 15.1.

Let $z \in A \cap C_x$. Then $Q_V z \in Q_V B(x, sh(x)/4)$, whence $|Q_V(x-z)| \leq sh(x)/4$. If $h(x) < |x-z|$, this gives $|Q_V(x-z)| < s|x-z|/4$, which means $z \in X(x, V, s/4)$ and so, by the definition of $h(x)$, $|x-z| \leq h(x)$. Thus $h(x) < |x-z|$ is impossible, and we have $|x-z| \leq h(x)$. Hence $|x^* - z| \leq 2h(x)$. Suppose $z \notin X(x^*, 2h(x), V, s)$. Then

$$\begin{aligned} s|x^* - z| \leq |Q_V x^* - Q_V z| &\leq |Q_V(x^* - x)| + |Q_V(x-z)| \\ &< s|x - x^*|/4 + sh(x)/4 \leq sh(x)/2. \end{aligned}$$

Using this and $|x - x^*| \geq 3h(x)/4$, we get

$$|x - z| > 3h(x)/4 - h(x)/2 = h(x)/4 \geq |Q_V(x-z)|/s.$$

Hence $z \in X(x, 2h(x), V, s)$, which proves (1).

From (1) and the hypothesis we get

$$\mathcal{H}^m(A \cap C_x) \leq 2\lambda(2h(x))^m s^m.$$

By the covering theorem 2.1 there is a countable set $S \subset A$ such that the balls $Q_V B(x, sh(x)/20)$, $x \in S$, in the m-dimensional plane $V^\perp$ are disjoint and

$$Q_V A \subset \bigcup_{x \in S} Q_V B(x, sh(x)/4).$$

This means

$$A \subset \bigcup_{x \in S} C_x.$$

Hence (recall $\mathcal{H}^m(V^\perp \cap B(y,r)) = 2^m r^m$ for $y \in V^\perp$)

$$\begin{aligned} \mathcal{H}^m(A) &\leq \sum_{x \in S} \mathcal{H}^m(A \cap C_x) \leq 2\lambda\, 2^m \sum_{x \in S} (sh(x))^m \\ &= 2^{m+1}\lambda\, 20^m 2^{-m} \sum_{x \in S} \mathcal{H}^m\big(V^\perp \cap B(Q_V x, sh(x)/20)\big) \\ &\leq 2 \cdot 20^m \lambda \mathcal{H}^m\big(V^\perp \cap B(Q_V a, \delta/2)\big) = 2 \cdot 20^m \lambda \delta^m. \end{aligned}$$ □

The following corollary says that a purely m-unrectifiable set is rather scattered; it approaches almost all of its points from all directions. Recall from Chapter 11 that sets of Hausdorff dimension bigger than m behave similarly.

15.15. Corollary. *If $V \in G(n, n-m)$, $\delta > 0$ and $A \subset \mathbf{R}^n$ is purely m-unrectifiable with $\mathcal{H}^m(A) < \infty$, then*

$$\limsup_{s \downarrow 0} \sup_{0<r<\delta} (rs)^{-m} \mathcal{H}^m\big(A \cap X(a, r, V, s)\big) > 0$$

for $\mathcal{H}^m$ almost all $a \in A$.

Proof. Let B be the set of those points $a \in A$ for which the assertion fails, let $\lambda > 0$ and define

$$A_i = \big\{a \in A : \sup_{0<r<\delta} (rs)^{-m} \mathcal{H}^m(A \cap X(a, r, V, s)) < \lambda \quad \text{for } 0 < s < 1/i\big\}.$$

Then $A_1 \subset A_2 \subset \ldots$ and $B \subset \bigcup_{i=1}^{\infty} A_i$. By Lemma 15.14 we have for $i = 1, 2, \ldots,$

$$\mathcal{H}^m\big(A_i \cap B(a, \delta/6)\big) \leq 2 \cdot 20^m \lambda \delta^m \quad \text{for } a \in \mathbf{R}^n.$$

Hence

$$\mathcal{H}^m\big(B \cap B(a, \delta/6)\big) \leq 2 \cdot 20^m \lambda \delta^m.$$

Letting $\lambda \downarrow 0$ we see that B intersects every ball of radius $\delta/6$ in a set of $\mathcal{H}^m$ measure zero, which yields $\mathcal{H}^m(B) = 0$. □

We give also a variant for a fixed s with a quantitative lower bound. Compare this with Theorem 11.10 for sets of dimension greater than m.

15.16. Corollary. *If $V \in G(n, n-m)$, $0 < s < 1$ and $A \subset \mathbf{R}^n$ is purely m-unrectifiable with $\mathcal{H}^m(A) < \infty$, then*

$$\Theta^{*m}\big(A \cap X(a, V, s), a\big) \geq 240^{-m-1} s^m$$

for $\mathcal{H}^m$ almost all $a \in A$.

Proof. The set of points $a \in A$ where this fails is contained in the union of the sets A_i, $i = 1, 2, \ldots,$ which consist of points $a \in A$ for which

$$\mathcal{H}^m\big(A \cap X(a, r, V, s)\big) < \lambda s^m r^m \quad \text{for } 0 < r < 1/i$$

with $\lambda = \frac{1}{3} \cdot 120^{-m}$. If $0 < \delta < 1/i$, Lemma 15.14 yields

$$\mathcal{H}^m\big(A_i \cap B(a, \delta/6)\big) \leq 2 \cdot 20^m \lambda \delta^m \quad \text{for } a \in \mathbf{R}^n,$$

whence $\Theta^{*m}(A_i, a) \leq 2 \cdot 60^m \lambda < 2^{-m}$. In view of Theorem 6.2 (1) this implies $\mathcal{H}^m(A_i) = 0$ and proves the corollary. □

Approximate tangent planes

We can now establish the relations between rectifiability and existence of tangent planes. As indicated in Example 15.1 we need to use approximate tangent planes rather than the ordinary ones.

15.17. Definition. *Let $A \subset \mathbf{R}^n$, $a \in \mathbf{R}^n$ and $V \in G(n,m)$. We say that V is an approximate tangent m-plane for A at a if $\Theta^{*m}(A,a) > 0$ and for all $0 < s < 1$,*

$$\lim_{r \downarrow 0} r^{-m} \mathcal{H}^m\big(A \cap B(a,r) \setminus X(a,V,s)\big) = 0.$$

We then write $\operatorname{ap}\operatorname{Tan}^m(A,a)$ for the set of all approximate tangent m-planes of A at a. If there is only one plane V in $\operatorname{ap}\operatorname{Tan}^m(A,a)$, we shall also write $V = \operatorname{ap}\operatorname{Tan}^m(A,a)$.

If $m = 1$, the approximate tangent line is unique, when it exists, but for $m \geq 2$ it need not be unique, see Exercise 4. It would not be difficult to show directly with the help of Theorem 6.2 that if A is $\mathcal{H}^m$ measurable and $\mathcal{H}^m(A) < \infty$, then the approximate tangent m-plane is unique at $\mathcal{H}^m$ almost all points of A where such a plane exists. But since we are going to get this information for free from the proof of Theorem 15.19, we shall not prove it separately.

15.18. Lemma. *Let A and B be $\mathcal{H}^m$ measurable subsets of $\mathbf{R}^n$ such that $B \subset A$ and $\mathcal{H}^m(A) < \infty$. Then for $\mathcal{H}^m$ almost all $a \in B$, $\operatorname{ap}\operatorname{Tan}^m(B,a) = \operatorname{ap}\operatorname{Tan}^m(A,a)$. (Note that $\operatorname{ap}\operatorname{Tan}^m(A,a)$ may also be empty.)*

Proof. This follows directly from Theorem 6.2 (2). □

15.19. Theorem. *Let E be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$. Then the following are equivalent:*

(1) *E is m-rectifiable.*
(2) *E is m-linearly approximable.*
(3) *For $\mathcal{H}^m$ almost all $a \in E$ there is a unique approximate tangent m-plane for E at a.*
(4) *For $\mathcal{H}^m$ almost all $a \in E$ there is some approximate tangent m-plane for E at a.*

Proof. That (1) implies (2) follows from Theorem 15.11. Using the facts that $\Theta^{*m}(E,a) \le 1$ for $\mathcal{H}^m$ almost all $a \in E$, by Theorem 6.2 (1), and that

$$B(a,r) \setminus X(a,V,s) \subset \big(B(a,r) \setminus V_a(\varepsilon s r)\big) \cup B(a,\varepsilon r),$$

one easily deduces (3) from (2); for the uniqueness use (15.8). Clearly (3) implies (4).

We are left to show that (4) implies (1). By Lemma 15.18 this is the same as showing that a purely m-unrectifiable set E fails to have an approximate tangent plane almost everywhere. This is fairly obvious from Corollary 15.16, but let us fill in the details. So assume E to be purely m-unrectifiable.

We can cover the compact space $G(n,m)$ with finitely many balls $B(W,1/3) = \{V : \|P_V - P_W\| \le 1/3\}$. Hence it is sufficient to show that for a fixed W the set B of those $a \in E$ for which $V_a \in \operatorname{ap}\operatorname{Tan}^m(E,a)$ exists and belongs to $B(W,1/3)$ has $\mathcal{H}^m$ measure zero. Suppose $\mathcal{H}^m(B) > 0$. Let $\lambda > 0$. Then for some $\delta_0 > 0$ the set C of those points $a \in B$ for which

$$\sup_{0<r<\delta_0} r^{-m}\mathcal{H}^m\big(B \cap B(a,r) \setminus X(a,V_a,1/3)\big) < \lambda\, 3^{-m}$$

has positive $\mathcal{H}^m$ measure. Since $\|P_{V_a} - P_W\| \le 1/3$, we have for $r > 0$,

$$X(a,r,W^\perp,1/3) \subset B(a,r) \setminus X(a,V_a,1/3),$$

because $|P_W(x-a)| < |x-a|/3$ implies $|P_{V_a}(x-a)| < 2|x-a|/3$ and further $|P_{V_a^\perp}(x-a)| > |x-a|/3$. Thus for $a \in C$,

$$\mathcal{H}^m\big(C \cap X(a,r,W^\perp,1/3)\big) < \lambda\, 3^{-m} r^m \quad \text{for } 0 < r < \delta_0.$$

Choosing $\lambda < 240^{-m-1}$ Corollary 15.16 leads to a contradiction. □

As usual, a characterization of rectifiability leads immediately to a characterization of pure unrectifiability; Lemma 15.18 and Theorem 15.19 yield the following.

15.20. Corollary. *Let E be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$. Then E is purely m-unrectifiable if and only if for $\mathcal{H}^m$ almost all $a \in E$,* $\operatorname{ap}\operatorname{Tan}^m(E,a) = \emptyset$.

We have proven the almost everywhere existence of approximate tangent planes for rectifiable sets, and we shall establish other properties later on, using Rademacher's theorem and Sard's theorem for Lipschitz maps. Another possibility is to involve deeper analysis to prove the following theorem and then use the properties of C^1 submanifolds and the density theorem 6.2 (2) to study rectifiable sets. For this see Federer [3, Chapter 3].

15.21. Theorem. *Let E be an $\mathcal{H}^m$measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$. Then E is m-rectifiable if and only if there are m-dimensional C^1 submanifolds $M_1, M_2, \dots$ of $\mathbf{R}^n$ such that*

$$\mathcal{H}^m\Big(E \setminus \bigcup_{i=1}^{\infty} M_i\Big).$$

Remarks on rectifiability

15.22. (1) Joan Orobitg has observed that one can replace $\mathcal{H}^m$ by $\mathcal{H}^m_\infty$ in 15.14–16 without any changes in the arguments. Since also the proof of the implication (4) $\Rightarrow$ (1) of Theorem 15.19 works for $\mathcal{H}^m_\infty$, the following statement holds.

If $E \subset \mathbf{R}^n$ is such that for $\mathcal{H}^m$ almost all $a \in E$ there exists $V \in G(n,m)$ for which

$$\lim_{r \downarrow 0} r^{-m} \mathcal{H}^m_\infty\big(E \cap B(a,r) \setminus X(a,V,s)\big) = 0$$

for $0 < s < 1$, then E is m-rectifiable.

Note that one does not have to assume that $\mathcal{H}^m(E) < \infty$, but as a consequence of rectifiability it follows that E has σ-finite $\mathcal{H}^m$ measure.

(2) The 1-rectifiability is often much easier to obtain than the higher dimensional rectifiability because of the following result: every compact connected set $K \subset \mathbf{R}^n$ with $\mathcal{H}^1(K) < \infty$ is a Lipschitz image of a subinterval of $\mathbf{R}$, see e.g. Falconer [4, § 3.2] and David and Semmes [2, Theorem I.1.8]. There is no analogue for higher dimensional sets, see Federer [3, 4.2.25].

(3) Jones [3] gave an interesting characterization of 1-rectifiability. Let $E \subset \mathbf{R}^2$ be compact. For any dyadic square Q let

$$\beta_E(Q) = \inf\big\{\delta > 0 : E \cap 3Q \subset L(\delta d(Q)) \quad \text{for some line } L\big\}/d(Q)$$

where $3Q$ is the square with the same centre as Q and $d(3Q) = 3d(Q)$. Jones proved that E is contained in some rectifiable curve if and only if

$$\beta^2(E) \equiv \sum_Q \beta_E(Q)^2\, d(Q) < \infty,$$

where the summation is over all dyadic squares. He also gave estimates for the length of the shortest curve containing E in terms of $\beta^2(E)$.

This leads immediately to characterizations of 1-rectifiable and purely 1-unrectifiable subsets of $\mathbf{R}^2$. For example, if $F \subset \mathbf{R}^2$ is $\mathcal{H}^1$ measurable with $\mathcal{H}^1(F) < \infty$ then F is purely 1-unrectifiable if and only if $\beta^2(E) = \infty$ for every $E \subset F$ with $\mathcal{H}^1(E) > 0$.

Jones's proof for the "if" part works also for one-dimensional sets in $\mathbf{R}^n$. Okikiolu [1] extended the more difficult "only if" part.

(4) Rectifiable sets occur as level sets for Lipschitz functions. Let $f\colon \mathbf{R}^n \to \mathbf{R}^k$ be Lipschitz with $k \leq n$. Then for $\mathcal{L}^k$ almost all $y \in \mathbf{R}^k$, $f^{-1}\{y\}$ is $(n-k)$-rectifiable, see Federer [3, 3.2.15]. That $f^{-1}\{y\}$ has σ-finite $\mathcal{H}^{n-k}$ measure follows already from Theorem 7.7. It was not known until recently whether one could say any more. For example when $n = 2$ and $k = 1$ the question was if for $\mathcal{L}^1$ almost all $y \in \mathbf{R}$, $f^{-1}\{y\}$ could be covered with countably many rectifiable curves (without the additional set of $\mathcal{H}^1$ measure zero). Konyagin [1] answered this negatively. He constructed a C^1 function $f\colon [0,1] \times [0,1] \to [0,1]$ such that for any $y \in [0,1]$, $f^{-1}\{y\}$ cannot be covered with countably many rectifiable curves.

(5) Rectifiability in more general metric spaces has recently been studied in Fremlin [1], Kirchheim [3] and Mauldin [1]. In Mauldin and Urbański [1] information is given about 1-rectifiability in connection with dynamical systems. M. Zähle [1]–[4] investigated integralgeometric properties and generalized curvatures for random and deterministic rectifiable sets. Anzellotti and Serapioni [1] investigated higher order rectifiability. Zaĭcev [1] studied tangential properties of sets with finite variations in the sense of Vitushkin. Relations between rectifiability, measure-theoretic boundaries and sets of finite perimeter can be found e.g. in Federer [3], Giusti [1], L. Simon [1] and Ziemer [1].

Uniform rectifiability

15.23. David and Semmes have developed an extensive theory of uniformly rectifiable sets, see David and Semmes [1]–[2] and the references given there. The existence of approximate tangent planes tells us that a set can be approximated by affine subspaces in small neighbourhoods. In uniform rectifiability this criterion for rectifiability is replaced by a quantitative notion, similar to that in Remark 15.22 (3) which gives information on how often such an approximation is good in dyadic cubes. David and Semmes also considered many other quantitative aspects of rectifiability including those which we shall study qualitatively in later chapters. We shall review some of these below. This study began from the attempts of David [1]–[3] and Semmes [1]–[3] to understand on what kind of m-dimensional sets singular integral operators are bounded in L^2 (see Chapter 20) and various square function estimates are valid. The first results on quantitative rectifiability were established by David in [1] and [3]. A problem in the calculus of variations related to uniform rectifiability was studied in David and Semmes [3]–[4], Dibos [1], Dibos and Koepfler [1] and Léger [1].

We shall now give a quick overview of some of the criteria for uniform rectifiability. A much more extensive survey is provided by the long and excellent introductory chapter of the book David and Semmes [2].

Let m and n be integers, $0 < m < n$, and let E be a closed subset of $\mathbf{R}^n$ such that for some constants $0 < c < d < \infty$,

$$(1) \qquad cr^m \leq \mathcal{H}^m\big(E \cap B(x,r)\big) \leq dr^m \quad \text{for } x \in E \text{ and } r > 0.$$

If $m = 1$, E is uniformly 1-rectifiable if and only if it is contained in a curve Γ which is regular in the sense that for some $C < \infty$,

$$\mathcal{H}^1\big(\Gamma \cap B(x,r)\big) \leq Cr \quad \text{for } x \in \Gamma,\ r > 0.$$

For higher dimensional sets the existing corresponding criterion is more complicated, involving maps which need not be Lipschitz; see David and Semmes [2, I (1.62)]. However, the following characterization can be given with Lipschitz maps, see David and Semmes [2, Theorem I.1.57].

E is uniformly m-rectifiable if and only if there exist positive numbers θ and M such that for each $x \in E$ and $r > 0$ there is a Lipschitz map $f\colon \mathbf{R}^m \to \mathbf{R}^n$ for which $\operatorname{Lip}(f) \leq M$ *and*

$$\mathcal{H}^m\big(E \cap B(x,r) \cap f(B(r))\big) \geq \theta r^m.$$

It is easy to verify from this that uniformly m-rectifiable sets are m-rectifiable. A corresponding characterization is valid also with bi-Lipschitz maps; for a result in this direction see also Toro [1].

The next condition is about approximation with planes, see David and Semmes [2, Theorem I.1.57]. It is an analogue of Jones's β^2-condition in 15.22 (3). It is also a manifestation of an analogy between rectifiability properties of sets and differentiability properties of functions in the spirit of the Littlewood–Paley theory, see David and Semmes [2, § I.1.3] for further discussions. See also Stein [1, Chapter VIII] and Dorronsoro [1] for such differentiability properties. Let for $x \in E$, $t > 0$,

$$\beta_1(x,t) = \inf_V t^{-m-1} \int_{E \cap B(x,t)} d(y,V)\, d\mathcal{H}^m y$$

where the infimum is taken over all affine m-dimensional subspaces of $\mathbf{R}^n$.

E is uniformly m-rectifiable if and only if there is $C < \infty$ such that for all $x \in E$ and $r > 0$

$$\int_0^r \int_{E\cap B(x,r)} t^{-1}\beta_1(x,t)^2\, d\mathcal{H}^m x\, dt \le Cr^m. \tag{2}$$

Conditions of this type are often called Carleson measure conditions. Note that (2) implies that often $\beta_1(x,t)$ must be small, whence there is a good approximation with m-planes. We shall also discuss other Carleson measure conditions and for this purpose it is convenient to define the notion of a Carleson set.

A Borel subset A of $E \times (0,\infty)$ is a Carleson set if there exists $C < \infty$ such that

$$\int_0^r t^{-1}\mathcal{H}^m\big(\{y \in E \cap B(x,r) : (y,t) \in A\}\big)\, dt \le Cr^m \tag{3}$$

for $x \in E$ and $r > 0$.

In a sense, Carleson sets are small subsets of $E \times (0,\infty)$.

To state another criterion for uniform rectifiability, given in David and Semmes [2, Theorem I.2.4], in terms of approximation with m-planes, define for $x \in E$, $t > 0$,

$$\beta(x,t) = \inf_V \big(t^{-1}\sup\{d(y,V) : y \in E \cap B(x,t)\}\big) \tag{4}$$

and its bilateral variant

$$\begin{aligned} b\beta(x,t) = \inf_V \big(t^{-1}(&\sup\{d(y,V) : y \in E \cap B(x,t)\} \\ &+ \sup\{d(z,E) : z \in V \cap B(x,t)\})\big) \end{aligned}$$

where the infima are again taken over all m-planes in $\mathbf{R}^n$.

E is uniformly m-rectifiable if and only if for every $\varepsilon > 0$ the set

$$\big\{(x,t) \in E \times (0,\infty) : b\beta(x,t) > \varepsilon\big\}$$

is a Carleson set.

The weaker condition where $b\beta(x,t)$ would be replaced by $\beta(x,t)$ is not sufficient even for rectifiability, see David and Semmes [1, § 20]. The above Carleson set condition can be considered as a quantitative version of the weak m-linear approximation property of 15.10. Hence the above characterization of uniform rectifiability quantifies the first part of Theorem 16.2 of the next chapter. Their proofs also have similarities.

If E can be locally approximated by m-planes, then for two points $y,z \in E$ close to each other the symmetric point of z with respect to y, that is $2y-z$, lies usually near E. This symmetry can be used to characterize uniform rectifiability, see David and Semmes [2, Corollary I.2.10].

E is uniformly m-rectifiable if and only if for every $\varepsilon > 0$ the set

$$\left\{(x,t) \in E \times (0,\infty) : \sup\{t^{-1}d(2y-z,E) : y,z \in E \cap B(x,t)\} > \varepsilon\right\}$$

is a Carleson set.

A similar symmetry condition appeared also in the proofs of Marstrand [3], Mattila [1] and Chlebík [1].

The following is another Littlewood–Paley type condition characterizing uniform rectifiability. It also provides a link to singular integrals which will be discussed in Chapter 20.

E is uniformly m-rectifiable if and only if for each compactly supported odd C^∞ function $\psi \colon \mathbf{R}^n \to \mathbf{R}$ there is $C < \infty$ such that

$$\sum_{k=-\infty}^{\infty} \int_E \left|2^{-km} \int_E \psi(2^{-k}(x-y))\, f(y)\, d\mathcal{H}^m y\right|^2 d\mathcal{H}^m x \le C \int_E |f|^2\, d\mathcal{H}^m$$

for all $f \in L^2(\mathcal{H}^m \llcorner E)$.

For this see David and Semmes [2, Theorem I.1.66]. Other criteria for uniform rectifiability will be discussed in 16.8 (3), 17.12 (3), 19.18 (5) and 20.29 (2).

Exercises.

1. Show that the set F of Example 15.2 does not possess an approximate tangent at any of its points.
2. Show that the set F of Example 15.2 is purely 1-unrectifiable without using the general theory of rectifiability.

3. Prove Lemma 15.5.
4. Show that the approximate tangent m-plane as defined in 15.17 is unique if $m = 1$, and need not be unique if $m \geq 2$.
5. Let $A \subset \mathbf{R}^n$ be $\mathcal{H}^m$ measurable with $\mathcal{H}^m(A) < \infty$. Show that at $\mathcal{H}^m$ almost all points $a \in A$ the following are equivalent:

$$(1) \qquad V \in \operatorname{ap\,Tan}^m(A, a).$$

$$(2) \qquad \lim_{r \downarrow 0} r^{-m} \mathcal{H}^m\big(A \cap B(a, r) \setminus V_a(\delta r)\big) = 0$$

for all $\delta > 0$ where $V_a = V + a$.
6. Show that the set C in Example 14.2 (3) is purely 1-unrectifiable.
7. Use Theorem 15.21 and the information that a C^1 submanifold has an ordinary tangent plane at all of its points to prove the implication (1) $\implies$ (3) of Theorem 15.19.
8. A set $S \subset \mathbf{R}^n$ is called an m-dimensional Lipschitz graph if there are $V \in G(n, m)$ and a Lipschitz map $g \colon V \to V^{\perp}$ such that

$$S = \{x + g(x) : x \in V\}.$$

Show that a set $E \subset \mathbf{R}^n$ is m-rectifiable if and only if there are m-dimensional Lipschitz graphs $S_1, S_2, \ldots$ such that $\mathcal{H}^m(E \setminus \bigcup_{i=1}^{\infty} S_i) = 0$. Hint: You can assume $E = f(\mathbf{R}^m)$ for some Lipschitz map $f \colon \mathbf{R}^m \to \mathbf{R}^n$. Split the set of those $x \in \mathbf{R}^m$ where f is differentiable and $\dim f'(x)\mathbf{R}^m = m$ into small parts A_i where $f'(x)$ approximates f uniformly and $f'(x)$ and $f'(y)$ are close to each other for x, $y \in A_i$.
9. Let F be a compact subset of $\mathbf{R}^2$, $\varepsilon > 0$ and $E = \{x \in \mathbf{R}^2 : d(x, F) = \varepsilon\}$. Prove that E is 1-rectifiable. Hint: Show that for each $x \in E$ there are r, θ and α such that $E \cap X^+(x, r, \theta, \alpha) = \emptyset$ with the notation of 11.10. Deduce the rectifiability of E from this.

Refinements of this result can be found in Oleksiv and Pesin [1].